

ActInf Textbook Group ~ Cohort 6 ~ Session 5

(Chapter 2, Part 2) ~ 2/26/2024

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all right welcome back cohort 6 Andrew however you want to begin this discussion on chapter two go for it sure um probably yeah pulling up the textbook might be a good move um I know that this is week two of chapter two so I don't want to uh reintroduce the entire chapters if you haven't already spent a week on it but I thought it'd be nice for anyone who wasn't able to attend last week to just kind of quickly skim what's going on um so yeah this is chapter 2 the low road to active inference in chapter one we were introduced to the idea that there's a high and low road to inference so um the low road is more about the um sort of the the the what or the the how of of active inference and um as opposed to the high road will where there will be more disc discussion on the why and the why of like why is this happening why are agents doing this how do they survive that's kind of the questions getting at here this is getting much more into sort of something like the mechanics of the process um and it picks up uh with the helm Holan perspective which is a 19th century late 19th century tradition uh that just began on viewing uh perception as unconscious Ence uh this idea is picked up more in ideas such as the Basi and brain hypothesis later on in the 20th century and from just an act uh active inference perspective we're treating not just perception but also action planning and learning all as basian inference problems as well as deriving a variational approximation um for these problems to overcome intractability problems um with the helm holian tradition the kind of Distinction about this versus other historical cognitive science Traditions is that rather than viewing perception is only a bottomup process of sensory States being fed into uh being turned into internal representations of the outside or some such thing like that instead perception is viewed as an inferential process that combines topdown prior information about the most likely causes of Sensations with bottom up sensory stimuli so we actually have this as opposed to um perception being more of an outside in process it's more of a it's it's both right it's bidirectional um we're introduced to baz's rule which is just um incredibly crucial to active inference as well as just many other kinds of probability problems I strongly recommend those

who are wanting to get into the maths to kind of work on um just kind of understanding baz's rule almost as like a sort of Cornerstone from which to begin um it can be viewed as a posterior uh equaling and uh a fraction that is uh the numerator is a prior and a likelihood and then your denominator is evidence and so what can happen is that you can update any of these Quant quantities in order to solve for another quantity right so this this this can be updated over time by taking in new information that then update everything else you can update your priors your your posteriors which are your beliefs at the end like at the end of the process um and so what's nice is that we can actually map all these quantities uh onto the modeling process and active inference it's so a generative model in our case like an agent um it could be a human being it could be uh some other biological organism or otherwise the generative model is just a combination of PR uh Prior PRI and um likelihood mappings so priors would be something like probability of X in the textbook um in often times uh notation can also be like uh probability of O which stands for or excuse me S which stands for latent States or hidden States um likelihoods are probability of Y given X so y condition on X or in other literature can be viewed as um observations o conditioned on S hidden Latent States so we have these other quantities um involved and yeah all of those are updated using bases rule um we have some additional um information in that section box 2.1 goes over gives a brief refresher on probabilistic reasoning some Rule and product rules for anyone who wants a little bit of a who's completely you know coming into this or or maybe took a probability course some but maybe you know it's uh the the knowledge is no longer no longer there and need a bit of a refresher um we have a nice example of of an agent who confronts um you know a frog or an apple and those are kind of two example like hidden States in that particular problem right so the agent has its own beliefs about the probability it will see a frog that it will see an apple uh it's given a new observation in the moment whatever the thing is it's looking at it jumps and so that's an observation versus the S the X the hidden states which are a frog or an apple and so those get ma those get linked together through um the the likelihood in baz's rule as we uh as we mapped a moment ago and so the agent basically updates its beliefs about what it's seen based on the observation it just took in not sure what the thing is it could be a frog could be an apple took in a new observation it jumped uh it has its own likelihood mappings oh if it jumped then uh excuse me um yeah then it will it'll draw inference from there so we're also introduced other Concepts like surprise basine surprise K Divergence um importance of biological plausibility um many of these theories are being being drawn from Neuroscience um and so there's there's emphasis on things like cognitive costs of computation

um there's only so much one's brain can do and so whenever you consider how many computations would be involved in you know any example beyond what we just looked at like two hidden States apples frogs um you can imagine and practice immensely more than that uh are being computed um presumably at any any given moment so um so that's where we end up looking at quantities like variational free energy uh as well as expected free energy by the end of the chapter um and this chapter I would say um this one's a really good Cornerstone for just like really getting familiar with a lot of what's going on in active inference it's it's a pretty lengthy chapter compared to the rest and um so spending some time with the different terms that are all kind of quickly being introduced here is a good idea so I guess I'll I'll kind of leave it off there and ask is anyone um you know curious about anything or more specific questions about what are going on um in in this chapter um it is a lot to say again yeah oh um so did the low road and high road come to get came did it come about like simultaneously or is it like is it just like U worked on like um like separately and then they both just came in at the same time um I do think uh Daniel would have uh a lot more knowledge on this but as far as I'm aware there's they're yeah these are coming from various traditions and are kind of a synthesis of different um yeah different Traditions or otherwise over time and then a lot of it is kind of uh composed primarily by Carl friston one of the co-authors of of the textbook yeah that's good answer a few other thoughts the rhetorical device or the metaphor was introduced by friston in this like beyond the desert landscape paper it's reference somewh theota um any model that exists is going to have a low road so to speak CU there's always a how otherwise it's not being it's just a a framework being discussed and then any kind of aaic starting point is its own kind of high road so low and high road can apply different to to to to probably to Frameworks other than active inference y um like kind of by analogy like to almost to denit cranes and skyhooks and like that kind of bottom up and top down Theory and empiricism so I think it kind of describes that generally and then the special thing about the low road uh constructions and that lead to active inference or like that are represented by the active inference gener model is exactly what Andrew said which is that the Bas rule is embodied totally by the generative model and that describes this like broad class of of constructible but then compatible with in this case free energy principle and not necessarily something else yeah I I guess my my thinking was you know did it came from two minds or was it from like one mind um using two different um perspectives to uh to think about you know um a solution to the problem I think it was many more than two overall like many people have raised many different good points about how something like has to be to persist for example and then

obviously many people have proposed different architectures and there's multiple ways to do active so it's kind of I think the you know maybe get towards higher resolution maps for the low on the high road and other things like that yep okay I thank you yeah David then Susan yeah so similar along those lines if you go back to uh the image 2.1 I think it was um between the percentage uh frogs and apples I think it was a comparison so in terms of that likelihood at what point are we talking about the same thing in terms of like if you have a belief structure of you know what percentage of apples believe they're frogs couldn't you wouldn't it be commutative in the same way where frogs would also have that same belief structure of apples if that makes sense in terms of you know occupying the same space so self- inferential are you a frog or are you an apple your prior still need to updated in terms of that correct but there would still be that percentage would that percentage not go away despite you know replacing those priors in that sense yeah that's very interesting question so in this case this likelihood model is the a matrix in figure 4.3 because it's describing the mapping between is an apple or and what is the observation so this is kind of an exteroceptive a Matrix however the metacognitive models treat like ascending observations as observations of metacognitive observers so in that case that could refer to like interceptive data coming in observ interceptive observations or introspective observations mapping to unobserved States about what kind of thing is generating those observations like am I hungry given the stomach sensation so it could be applied internally as well to maybe some of the things you're talking about yeah I guess I was speaking more also in terms of uh what how just said in terms of uh you know of one mind or of you know more than two minds as you had mentioned okay thanks Susan yeah so as far as um describing the low road and the high road would it be incorrect to um use a colloquial term like Tactical for low road tactical moves does anyone think yeah go for it yeah it's an interesting question um I mean you know on the one hand I'm not so I'm not so sure if if that can you know be answered expressly maybe that's the kind of thing that I think someone might have to kind of do their own write up on on justifying that as kind of maybe like part of the active inference ontology is to view it as tactical um I mean if you were thinking to just the context just within the confines of this chapter um and you're thinking about an agent being tactical um I would say this stuff relating actually to maybe planning that we see in later sections um that might be more directly linked just conceptually with the word tactical in the sense of an agent is comparing potential sequences of actions or policies that it could take um it is scoring those based on the expected free energy for each policy uh you end up going with the one that has the lowest expected free energy right so that's a kind of like what should

I do here a series of paths I could take here's the one that is least surprising or at least is expected to have the least amount of free energy returned take that path right so it's almost like strategizing now that that itself is still kind of a loose analogy and that you know yeah yeah I can see that yeah so let me ask you another question that may relate it relate it one one thought Susan on that um Andrew that's that's a good answer like doing the evaluation and the the search is H is the how and then there's other attributes of planning that would make up like kind of the high road of planning and I think the stream to look at is 42 um so this is also very useful for nested modeling and understanding the composability of the nested modeling so here it was like analogy to strategy tactics control the center as being a higher order simpler less tactical less operational side and then their model of a robot in a warehouse has this finer scale that is like tactical that describe the posture of the robot and then a higher coarser scale that was like the location of it now this model itself could be interpreted from a low road perspective because even though this higher order is about the location of the robot it is still constructed so there's probably a lot to look at it or to think about it especially if you worked it out actually I think I think you got strategic and tactical on both ends of action and perception I mean that's what kind of came to mind when he was he was talking and and and so and I don't know if this relates directly or not but when we talk about Li high road and low road policies that do they do they have distinctly different policies no they're meeting an active inference because you only build the active inference generative model that's actually doing the inference so they just come together in that constructed model yeah I think if it's useful um I've almost wanted to bring this up from an earlier question but um figure 1.2 in the chap in the textbook which is actually in chapter one page seven um this is just to illustrate like with active inference in the center right like these are the high and low roads to active inference like it's not so much that they're entirely distinct you know things it's more about conceptually these are two paths that you could kind of take to kind of move towards standing active inference right so it's like the low road is kind of we're starting from base theorem we're starting from statistics numbers calculation valuations so on and that's what this chapter is about meanwhile the high road is more about the free energy principle so now we're more expressly talking about things like like with perception everything we've talked about so far and scene chapter 2 we don't really see so much actually about like an agent trying to survive right we don't we don't quite see that this is just like math right like it's just doing stuff um not so much like trying to it's only near the end that we get mentions of like it has preferences that go into planning and things like that but really this is just kind of the mechanical aspect it's

yeah the high Rogue will be more it'll give us much more context and I also think it makes active inference so much more interesting when trying to think about how how to apply it to different situations so yeah H so I guess one of the purposes of active influence is so that we don't have to manually choose the priors right choose prior or choose what because in I guess in real life with organisms you know there's uh many fires hidden fires that you know we we don't know that lead to you know the perceptions and decisions that you know they they have um you know and it'll be too much for us to to try and emulate or simulate um I mean do we still have to generate the pr beforehand for an agent so that is something about active inference that you know there's there's been many conversations about this and you know it can it could totally Branch out into much bigger philosophical questions but active inference doesn't necessarily um have any guarantee of what a a prior should be it doesn't you know it's not very good whenever it comes to origin stories um I I suppose is one way of stating it it's more about this is how things happen this is why things happen I will say um if you're coming from more of like a computational modeling or coding perspective uh and you'll see this much later in the textbook in like chapter 7 for example um but oftentimes what's done is you'll establish some kind of vector of priors over initial States um so you do kind of have to hardcode those um the thing is you can you can have the agent you know interact with its environment running trials and then it can in fact from trial to trial update its own priors right so so it's kind of like once you get the the motion moving it it kind of moves on its own so to speak furthermore a lot of examples whenever it comes to modeling many people just start with uniform priors like there's no there's not something there necessarily that says you have to have you know some very um you know domain knowledge specific priors or or something like unleash the agent with uniform priors it doesn't know what's going on presumably it will you know as the low road mechanics play out it will it will readjust its own priors it kind of depends on the use case is the simplest answer yep so so some um so it can um generate it on price you know if it's given um you know like observations of like RGB you know an RGB camera and maybe yeah I don't know I guess it's just some sort of you know sensory input sure yeah it it is not widely or simply available that just because like you made a active influence agent is going to do arbitrary like infinite generalization structure learning Etc like those are very exciting directions to consider the kind of beginning State and the procedures for agents that could do things like modify the number of observations they received or understand different policy spaces so yep that's definitely to be explored and then play around with with the different components of that separately and then bring them together and see H how things go but if

you just look at like the PDP tutorial that is not not going to do like open-ended structure learning whereas maybe a certain kind of rapper around the recent structure learning paper would have that kind of capacity but then not necessarily like affect or attention or something else that's explored in different paper so it's always very situational which capacities are actually brought in Because unless there's like some kind of open-ended or metaprogramming element it's not going to do something that is not described by the generative models form is is that like a a big goal of active influence because I I mean it feels like you without that it would just be nearly impossible to to emulate you know the same fires that you living organisms would have because you we organisms would have a lot of fire like almost you know from our perspective almost infinite amount of f yes and depending on nich it's um but definitely some people that's similar to what they might be interested in like developing generalization capacities through metacognitive um nesting or through observers with higher complexity that might be something kind of related to how Wolfram is approaching it like with the Observer Theory and all of that and like computational capacity David yeah just an observation uh can you go back to the uh the branch map that you had with um basian and fren was [Music] uh was related to what Susan was asking it had a Bas them at the bottom and it had um active inference at the top I believe was a tree tree map figure one with the high and low road yes okay yes all right yes so sorry so yeah Fe so I guess just like visually I construct it this way where could you consider this if you linked base with free energy principle that that would you know cover a surface so in that case if you could do that would you have Baye over free energy or would it be free energy over Baye or would it be commutative if uh you can see the structure there maybe write it down like write it out as a question for the chapter and let's return to it but okay it sounds interesting yeah it's outside of the bounds of what we're covering so thanks it's just hard to process in real time but it sounds interesting okay yeah anyone else like to we can go totally any Direction with two or look ahead or look to some other part one more thought on the um question about where like the priors come from this is coming up later in chapter nine but chapter nine is about integrating real data so that kind of addresses some of the questions that come up about how to integrate with real systems because like it seems like pretty clearly one kind of thing to just write out a model or like sketch it out and just propose it and then it's another thing to connect it to data and and use it in like empirical settings so here's kind of that model that is constructed of like the rat and the Tas and then but that's a map of the rat's cognitive territory not claiming to be a total description of the rat's physiology just the team's decision as analyzed a certain way um so it's the Observer or the experimenter

who's passing what become observations to the rat but they're actions from the scientist and then the emitted actions of the Rat are observations for the scientist so it is possible to understand like our role around like building the generative model in terms of this kind of laboratory setting kind of consistent with the origins and the behavioral analysis and SPM like in a laboratory making a cognitive map then that map is a map of environments and then that that was kind of explored in that in a really clear and interesting Way by by this paper with ramstead Sak divid devel and friston I'll put in the live chat but they they they make very strong claims like about that kind of relationship essentially saying if this is how you see the situation then Fe is the best descriptor of it so people could descent from their conclusions or from their um setup of the problem or or features but it's a very strong claim within this setting and then the the fallacy fallacy like the map territory fallacy is very ly discussed like the map of San Francisco is not the actual land or the territory of San Francisco and there's like the Bor his story about the Ono one map and how that becomes not useful because it's kind of like wrapped over the real world so maps are always kind of motivated by the right um course gradings and then um so that's the map territory fallacy is falsely or inappropriately articulating the map territory relationship but then they are resp responding to what they perceived is a criticism of O Fe which is um once it's acknowledged as a cognitive modeling technique then people say well then it is just a map we want the real territory description and we want the real thing so then that's like the fallacy fallacy is their kind of term for the overshoot it's like okay well then it doesn't describe anything because you can just construct whatever map you want it's like yes you can construct whatever map you want of San Francisco and then what so that's kind of interesting angle that they brought out even a few years ago David yeah so in terms of that I'm just trying to think of how to how to rephrase it um terms of the mapping itself I mean obviously like through periodic time change and everything but if you created the map itself like if you I'm thinking in terms of like wave function collapse right if you're if you're observing something directly it will collapse the way it function just you know as a basic overview of it but if it's oblique to your perspective but it's still within your it's not necessar necessarily in your priors or it might have be been dissolved from your prior into something else that's kind of your own inferential map which you're not collab but you're utilizing without having to you know use that extra energy or any extra energy Beyond you know just that distilled information that you have that's kind of how I reinterpreted that that sounds very interesting like there's like a primary sensory mapping that's like dealing with a collapse over like primary observations or something but then right so

instead of like a singular Viewpoint of the map and all of its other points the Viewpoint comes from you know a different perspective that you're not particularly aware of at the time of your inferential connection to the map itself so the map still exists but you're not using any energy to look at it directly so it kind of mercades itself over your own perspective okay cool add more but it sounds interesting Cheryl okay yes hi um I've been um as you know I'm new to this but I am very much interested in the way cognition is defined so I'm my first question is does the AC of inference then just focus on observable behaviors because so much of what the brain does is implicit and is not going to actually show up as a something that an outsider could even observe so I'll stop with that question before I go on it's a great question so I I wouldn't say it models um only observable behaviors but rather it is made clear what is observable and what is not so you can make a model that's fully observable like a chessboard where there is no difference between the observations in the hidden State like of the piece on the board so it's like a fully observable system or you can have a system where the only observation is um a binary State and then there's like very complex internal Dynamics and another angle on that um is from this paper with bongard L so here they talk about this like poly Computing which is that the Observer is doing inference on what kind of computation a material is doing so if like there was an encrypted communication to an observer who didn't know it would look like noise that's why the observer's um situatedness is so important because we we shouldn't just assume just because it was noise to one Observer doesn't mean that there isn't something in that information stream but it was their relationship of it that it was perceived to be doing just noise generation whereas in a different reference frame that could have other information that can't be ruled out even from from any kind of noise pattern so like I would say it's it's a big philosophical and empirical question how you model like the priors and the the hyper priors and all this of observers and that just like empirically there are techniques that basian researchers use to address some of these things like testing the the influence of different prior distributions or building the initial priors from empirical data but certainly there's a lot of expressivity and probably a lot of complexity to to what you're asking about right okay that that's helpful so um active inference does not claim to be the way that this works but a way yes that's totally true it's totally true that's why it's often helpful to um differentiate between like when are we talking about like active inference as is versus with how it can be which is improving and learning active inference or active inference versus an alternative account of the same phenomena something that like Ryan Smith and other researchers do do and and hopefully will happen more that's one kind of comparative mode

which is a totally valid and important mode because it's important um for certain questions and situations and everything like to know like does reinforcement learning fit better than active inference in this setting there other times there's kind of epistemic value even with just developing along just an active model but is always useful to have even really radically different models like is this predictor even better than just a coin flip and then that helps calibrate it to to like more reference points than just parameter sweeps within the otim model which then doesn't allow you to make points about that in relationship something that wasn't actually explored um so I just have a question about where to put questions um because you're right I think I need to write this out to get a better idea of what it is that I'm even asking you um but it has something to do with the difference between implicit and explicit so uh where would I where would I kind of think this through on right you could add a new row and like add it to this table or you could like literally just add it anywhere and okay and then do at like active inference Institute and then it it's just can be added anywhere you type it or call yeah okay great all right thank you but yeah these are interesting questions like does the implicit and explicit map on to observations and hidden States correct yeah and then is there another implicit and explicit for modeling is that the same one described by the internal by the hidden States and by the um observations not necessarily right because at one point I'm going to ask you about neur uh neurons and neural development and the the way that the brain actually um in in in embryos grows together mation how the these things it doesn't seem like a basian thing um and uh because they don't all come out different in and you know in a in a normal curve you you know most human beings have their brains developed along the same lines and so um I I just would like to know how active inference what what part does it play as you go down levels of analyses or way up in levels of analyses past just um agents all all great questions on the development line like the canalization of development like the regularity across individuals that's definitely a big topic like Mike Levin and other researchers especially have looked at that like Wasington epigenetic landscape and understanding how like cells come to have morphology and then I mean you brought up other things as well but yeah those are like those are all great great questions all right thank you so much any anyone else just go for it yes go for it um I thought we have a section for cohort six questions um yeah and yeah when I was waiting for the textbook I I do like to put down you know my thoughts in there as well oh awesome yeah let's look at let's look at some right now they do all go to the same underlying thing but if you wanted to be tagged with cohort 6 thank you for suggesting yeah put it in here and I'll make a chapter 3 one okay yes what happens if you just update the prior with the

posterior instead of using K Divergence um do you mean like what happens if you update your belief just like to last observation super simply instead of calculating anything more complex than that yeah without using the supplies and you know I guess this to me kind of feels like like a learning rate kind of thing um is it is it like that yeah like um it's also related to Colman filter and smoothing so it's like if we're um getting noisy thermometer measurements I mean we could just update our belief just to the noisiness of the thermometer measurement and just treat it like perfect measurement of a underlying process but the Dynamics of the underlying process like we might be able to get a way better model by just saying well it's a sinusoid hidden State and then we're just sampling from it so yeah I I guess it makes sense from that perspective but from the perspective of you know whether it's a fog or an apple if you see an apple jump it kind of feels like oh like it's um much it could be a much stronger update that oh it is not an apple it's actually a fog now yeah like in this world there's a 1% chance that apples jump so you're totally right that if this was a apples never jump like a heris encoded as um a column in this Matrix yeah then it would go 100% so it's like in practice you can do things like that that truncate or just like start with no noise on a certain thing y um Zach data yeah great question it's um one that also came up actually earlier so look at the cohort 5 um video on like reinforcement learning but the key difference is that in active inference we're directly talking about the joint distribution and the likelihood of the generative model and preferences so pragmatic value the way that they come into with the picture in decision making is by checking the difference between the preferences and the observations coming in whereas in reinforcement learning there's a secondary reward distribution that's proposed on top of the observation so like 37 degrees body temperature is rewarding and then you pursue rewarding policies versus in active inference 37 is expected and preferred C and then likely policies are sampled that doesn't mean any given active model is as good or better or anything at all at all so it's totally situational which one is more useful which one you're more comfortable with and so on but that's the difference I heard someone Susan said I heard someone say active is based on causality not correlation well you can use it to study correlations amongst variables you could also use it like a basian causal draft to study so-called causation amongst variables so then you could engage in the whole is the causative map the territory's causation that's a whole area like Mel Andrews and others have explored some of that philosophy um and Judea Pearl kind of and and other people delman in the statistical causal modeling with um Dynamic causal modeling like if there's a causal relationship identified in the hidden State between two brain regions in the fmri study does that mean they have a causal relationship like those are

questions that there's a lot of discussion on David yeah along those lines how would that relate to like a co- causal State co- causal State um could be like here there's two distributions here's temperature and humidity and they both and then the output is the D point so two things that jointly cause a third thing right I'm also thinking in terms of like a third that causes still a singular in terms of like a zor if you were thinking in terms of like a gated where it's not a choice but rather it's you know you have two individuals making separate decisions one person is the patient in the you know the case we're talking about here right terms of the patient being doing EEG scan or something with the doctors priors you know at you know going down to that 0.01 right that 1% Apple frog you know at what point is it co- causal then does it become a zor or does it become still in or state or joint co-state very interesting I think look at this guest stream 67 if you haven't which talks about like noisy independent experts and about how like with sensor arrays that having more noisy uncorrelated which is to say noisy components whereas like if all the experts agreed on a test you'd actually have low Precision relative to one with an uncorrelated error profile okay perfect I'll check that out thank you yeah someone else had their hand raised um okay just to clarify we want variation variational free energy to be as small as possible approaching zero or negative Infinity um in practice approaching some number value that's in arbit units um there was like a discussion on the Discord General about this that people could look at but like the free energy values itself is going to converge to some situation specific minimal value just like if you were doing least squares like sum of squares minimization for linear regression the best fit regression would converge to like sum sum of squares so just a a number sum across the squared error differences from the linear regression but then that like value is depending on the number of data points and the units and all this other like random stuff not really random stuff but it does depend on those things so um that's why Delta free energy and the gradient of free energy is used and all of the physics grounding is actually about the gradient around the free energy and that's why it's like incrementally optimizable because there's a surface that gradients on free energy can be taken on so it's like a landscape like a free energy landscape there's an elevation at that point that's like the actual score of like q and given Y and then there's the local slopes that are used for optimization and then the simplest optimization here is like is just go in the direction that it's going to be lower but that doesn't really matter what the absolute value is and then when it's when the change in free energy is zero the model is like converged to a stable Point could be local could be Global that's the whole challenge with like model yeah this is a great question I think maybe some people have put things people outside of active inference

have written about like so-called conspiracy theories and Bais and belief updating um also definitely some of these topics people have looked at like people can add links but let's continue all the other questions um generating priors I mean in practice they're specified by the person who's making the generative model there should be many priors and policies yeah to make the quote Whole iguana so this is in chapter 10 but chapter 10 is a lot like chapter one so um this is like in contrast to the research program of like let's study which which is also a very useful research program but kind of two two paths diverging one of them is like studying really narrow components like let's study mechanotransduction in this brain region of groph fruit fly and then understand very narrow features very deeply um but scattered across like the animal and and non-animal species and then in contrast like this um alternative holistic approach is model the whole uana which is like how about not just super good on the homeostasis of temperature on the Iguana then we know nothing about this other component but like think totally about it so yeah it is very challenging to make multimodal models because it's easy to make a single like an ant colony model with just foraging is easy but then if it's like foraging and Nest construction it's hard to do that David yeah so in terms of that wouldn't it wouldn't that come back then to context free grammars and being able to um compute better in that context because at what point you know when you start using frog iguana Apple I mean isn't that giving context to the measurement itself to begin with yeah very interesting question I think in the pedagogical context you could replace it with like any other it could be the the whole onion so like in that sense it doesn't like it's not a one specific just to be like clear um but like yes the reason why that makes sense or how somebody would know what to do is reflective of their ref I guess I guess it's kind of how I'm seeing it is in terms of like if you remove all the context from everything you're if you have the statistics remaining you could apply it to you know your own inference in terms of you know seeing that statistical Dynamic amongst two other things that or name something else you know that have the same exact n values that might end up you know then being if you remove the context from it you know the statistical correlation still exists yeah there's there's a ton of things to say about that I mean there are priors I suppose that you're bringing into that as well cool let's explore it more but yes it sounds interesting all right thanks Thomas for the live streams on the noisy experts that was 67.1 I think 67.1 ntqr Susan asks is degrees of freedom computed in act INF and not reinforcement that is not the case degrees of freedom are in place anytime statistics is being used so even if you were to use linear regression or reinforcement learning or neural network or llm Transformer any of those statistical settings degrees of freedom are in play Anova test

every single statistic setting with the meaning of what they describe the degrees of freedom of not sure if I'm using the right term are these equations proven either mathematically or empirically or are they useful approximations it's a complex question like in one sense all of the consistent spaces in math sub subspaces and so on like by virtue of the equal sign they're all describing topologies all of them could be reduced to like 1 equals 1 which could come back to like the axioms of how we are going to be defining like numbers and equality so there's an intrinsic logic to the equations that describes like the fact that it really is possible as um for example uh yakup and some other people have done like with equation 2.5 you really can go and look how the different decompositions really of free energy and expected free energy really are related to each other formally so there's an internal consistency similar to the status of all other like mathematical statistics or like mathematical physics does that provide an empirical validity is a totally interesting question it certainly does not provide situational validity or adequacy but then it's like a different interesting question does it like what are we looking for we say empirically and like the live stream from just a few days ago with Ryan Smith um on the empirical status is super relevant as I'm sure the insights that is going to be like later in the week with Ryan Smith so like and he talked a little bit about like his philosophy background in this stream it was really cool because most of his work is very clinical and like has to do with with SC like scientifically clinical empirical okay any last question so this was um chapter 2 low road the what slow chapter 3 kind of refreshes and picks up on the high road with uh more about the like transcendental imperatives side that are not necessarily Arch architecture dependent so I think like in that um the robotics work of JF clader at The Institute he's used symbolic programming to develop kind of an alternative low road that's not distributional belief based but it has the same high road imperatives so just kind of an example of how there can be like the same why um same Tios same um ends could be done with different low roads kind of interesting topics that like come out through chapter three then having approached actim from the two sides chapter 4 goes into the formal details of the generative model that's the central artifact that's like what is constructed and then chapter 5 gives a bunch of studies and reviews on one of the systems that's been the most studied which is like the mammalian nervous system so kind of just showing here's what constructing system specific models and linking them up looks like and and here's what it looks like in the several decades of study that people have done in the mammal nervous system just as kind of like a review SL jumping off point any last SL comments or thoughts H um I watched a video um with do uh Dr Jeff Beck um and I'm not sure if I'm remembering this

prop but they were saying there's some people trying to implement the Transformer architecture within active influence like does that make sense at all or maybe I'm just remembering wrong there's projects that are doing everything from like active rappers around Transformers to Transformer rappers around llms to Transformers or neural networks replacing specific matrices in generative models like using a neural network for a instead of just a matrix with the dimensions of observations by hidden States so there's many possible combinations I don't know what backck discussion it was but um he did talk with Darius and they like talked about markall blanket and brought up like a lot of useful questions and ments on this does that address it cool cool yep right thank you bye